

# Revisit decimal calculation

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: Calculate  $34.75 + 8.906$ . Copy or complete the first step.

2. Calculate  $50 - 17.485$ .

3. Calculate  $6.4 \times 7$ .

4. Miro says: Miro lines up the final digits rather than the decimal points. Is this correct? Explain.

# Teacher answers and checking prompts

1. Read the model: Calculate  $34.75 + 8.906$ . Copy or complete the first step.  
**Answer** 43.656.
2. Calculate  $50 - 17.485$ .  
**Answer** 32.515.
3. Calculate  $6.4 \times 7$ .  
**Answer** 44.8.
4. Miro says: Miro lines up the final digits rather than the decimal points. Is this correct? Explain.  
**Answer** Align decimal points so ones, tenths, hundredths and thousandths stay in their columns.

# Revisit decimal calculation

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Calculate  $34.75 + 8.906$ .

6. Check this result using an inverse, estimate or known fact:  
Calculate  $6.4 \times 7$ .

2. Calculate  $50 - 17.485$ .

7. A number divided by 8 is 3.75. What is the number?

3. Calculate  $6.4 \times 7$ .

8. Identify and correct this misconception: Miro lines up the final digits rather than the decimal points.

4. Solve this independently and show every stage: Calculate  $34.75 + 8.906$ .

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate  $50 - 17.485$ .

10. Write and solve a similar question to: Calculate  $6.4 \times 7$ .

# Teacher answers and checking prompts

1. Calculate  $34.75 + 8.906$ .  
**Answer** 43.656.
2. Calculate  $50 - 17.485$ .  
**Answer** 32.515.
3. Calculate  $6.4 \times 7$ .  
**Answer** 44.8.
4. Solve this independently and show every stage: Calculate  $34.75 + 8.906$ .  
**Answer** 43.656.
5. Use a second method or representation for: Calculate  $50 - 17.485$ .  
**Answer** Expected result: 32.515. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Calculate  $6.4 \times 7$ .  
**Answer** Expected result: 44.8. A valid check must be shown.
7. A number divided by 8 is 3.75. What is the number?  
**Answer** 30.
8. Identify and correct this misconception: Miro lines up the final digits rather than the decimal points.  
**Answer** Align decimal points so ones, tenths, hundredths and thousandths stay in their columns.
9. Write a complete sentence explaining how you checked one answer.  
**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Calculate  $6.4 \times 7$ .  
**Answer** Answers vary. The new question must use the same mathematical structure and include a correct solution.

# Revisit decimal calculation

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. A number divided by 8 is 3.75. What is the number?

6. Create a new example that tests the same idea as: Calculate  $6.4 \times 7$ .

2. Prove the answer to this question, rather than only calculating it: Calculate  $34.75 + 8.906$ .

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate  $50 - 17.485$ .

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 44.8.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro lines up the final digits rather than the decimal points.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

# Teacher answers and checking prompts

1. A number divided by 8 is 3.75. What is the number?  
**Answer** 30.
2. Prove the answer to this question, rather than only calculating it:  
Calculate  $34.75 + 8.906$ .  
**Answer** Expected result: 43.656. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate  $50 - 17.485$ .  
**Answer** Expected result: 32.515. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 44.8.  
**Answer** One valid reconstruction is: Calculate  $6.4 \times 7$ . Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro lines up the final digits rather than the decimal points.  
**Answer** Align decimal points so ones, tenths, hundredths and thousandths stay in their columns.
6. Create a new example that tests the same idea as: Calculate  $6.4 \times 7$ .  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.  
**Answer** Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.  
**Answer** Answers vary. The explanation must distinguish the mathematical structure from the changed value.